

Machine Learning Mini-Project

Problem Definition

The objective of this project is to develop a machine learning model that predicts whether a loan application will be approved or rejected based on applicant details.

Financial institutions receive many loan applications, and manually evaluating each application takes time and effort. This system helps automate the loan approval process by analysing applicant information such as income, loan amount, education, CIBIL score, assets, and employment status.

The model predicts the loan status as:

- **Approved**
- **Rejected**

Dataset Description

The loan approval dataset is a collection of financial records and associated information used to determine the eligibility of individuals or organizations for obtaining loans from a lending institution. It includes various factors such as cibil score, income, employment status, loan term, loan amount, assets value, and loan status.

<https://www.kaggle.com/datasets/architsharma01/loan-approval-prediction-dataset>

Model Building and Evaluation

The model is built using Decision Tree Classifier. This was selected for the project because the problem involves classification, where the objective is to predict whether a loan application will be approved or rejected. Decision Tree is suitable for classification problems as it makes decisions based on conditions and creates a tree-like structure for prediction.

Code:

```
import pandas as pd
import numpy as np
import joblib
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (accuracy_score, mean_squared_error)

data = pd.read_csv("Downloads/loan_approval_dataset.csv")
data.columns = data.columns.str.strip()
```

```

print('Data=',data.head())
print("\nDataset Shape Before Removing Duplicates:", data.shape)

data["education"] = data["education"].str.strip()
data["self_employed"] = data["self_employed"].str.strip()
data["loan_status"] = data["loan_status"].str.strip()

data = data.drop_duplicates()

print("\nDataset Shape After Removing Duplicates:", data.shape)

loan_status_encoder = LabelEncoder()
education_encoder = LabelEncoder()
self_employed_encoder = LabelEncoder()

data["loan_status"] = loan_status_encoder.fit_transform(data["loan_status"])
data["education"] = education_encoder.fit_transform(data["education"])
data["self_employed"] =
self_employed_encoder.fit_transform(data["self_employed"])

print("\nEncoded Dataset:")
print(data.head())

X = data.drop(['loan_id', 'loan_status'], axis=1)
y = data["loan_status"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test =
train_test_split(X_scaled,y,test_size=0.2,random_state=42)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print("\nRMSE:", rmse)

```

```

joblib.dump(model, "loan_approval_model.pkl")
joblib.dump(scaler, "loan_scaler.pkl")
joblib.dump(education_encoder, "education_encoder.pkl")
joblib.dump(self_employed_encoder, "self_employed_encoder.pkl")
joblib.dump(loan_status_encoder, "loan_status_encoder.pkl")

print("\nModel trained and saved successfully!")

```

Output:

```

PS C:\Users\hp> & C:/Users/hp/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/hp/Desktop/mlai/minipro.py
Data=
 loan_id  no_of_dependents  education  self_employed  income_annum  ...  residential_assets_value  commercial_assets_value  luxury_assets_value  bank_asset_value  loan_status
0      1      2      Graduate      No      9600000  ...      2400000      17600000      22700000      8000000      Approved
1      2      0      Not Graduate      Yes      4100000  ...      2700000      2200000      8800000      3300000      Rejected
2      3      3      Graduate      No      9100000  ...      7100000      4500000      33300000      12800000      Rejected
3      4      3      Graduate      No      8200000  ...      18200000      3300000      23300000      7900000      Rejected
4      5      5      Not Graduate      Yes      9800000  ...      12400000      8200000      29400000      5000000      Rejected

[5 rows x 13 columns]

Dataset Shape Before Removing Duplicates: (4269, 13)
Dataset Shape After Removing Duplicates: (4269, 13)

Encoded Dataset:
 loan_id  no_of_dependents  education  self_employed  income_annum  ...  residential_assets_value  commercial_assets_value  luxury_assets_value  bank_asset_value  loan_status
0      1      2      0      0      9600000  ...      2400000      17600000      22700000      8000000      0
1      2      0      1      1      4100000  ...      2700000      2200000      8800000      3300000      1
2      3      0      0      0      9100000  ...      7100000      4500000      33300000      12800000      1
3      4      3      0      0      8200000  ...      18200000      3300000      23300000      7900000      1
4      5      5      1      1      9800000  ...      12400000      8200000      29400000      5000000      1

[5 rows x 13 columns]

Accuracy: 0.977751756440281

RMSE: 0.14915845118436624

Model trained and saved successfully!
PS C:\Users\hp>

```

Accuracy=0.977751756440281

RMSE=0.14915845118436624

The **accuracy score** indicates how well the model correctly predicts loan approval and rejection. A high accuracy value suggests that the model performs well in classifying applicants based on the provided features.

The **RMSE (Root Mean Square Error)** measures prediction error. A lower RMSE value indicates that the model's predictions are closer to the actual results.

The obtained **accuracy of 0.9777 (97.78%)** indicates that the model performs very well in predicting loan approval status. This means that approximately **98 out of 100 predictions** made by the model are correct, showing high reliability in classification.

The obtained **RMSE value of 0.1491** is relatively low, which indicates that the prediction error is minimal and the predicted values are very close to the actual values. A lower RMSE generally represents better model performance.

Prediction Code:

```
import joblib
import pandas as pd

model = joblib.load("loan_approval_model.pkl")
scaler = joblib.load("loan_scaler.pkl")

education_encoder = joblib.load("education_encoder.pkl")
self_employed_encoder = joblib.load("self_employed_encoder.pkl")
loan_status_encoder = joblib.load("loan_status_encoder.pkl")

no_of_dependents = int(input("Enter number of dependents: "))
education = input("Education (Graduate/Not Graduate): ").strip()
self_employed = input("Self Employed (Yes/No): ").strip()
income_annum = float(input("Annual Income: "))
loan_amount = float(input("Loan Amount: "))
loan_term = int(input("Loan Term: "))
cibil_score = int(input("CIBIL Score: "))
residential_assets_value = float(input("Residential Assets Value: "))
commercial_assets_value = float(input("Commercial Assets Value: "))
luxury_assets_value = float(input("Luxury Assets Value: "))
bank_asset_value = float(input("Bank Asset Value: "))

education = education_encoder.transform([education])[0]
self_employed = self_employed_encoder.transform([self_employed])[0]

new_data =
pd.DataFrame([[no_of_dependents,education,self_employed,income_annum,loan_amount,loan_term,
cibil_score,residential_assets_value,commercial_assets_value,luxury_assets_value,bank_asset_value]],
columns=["no_of_dependents","education","self_employed","income_annum","loan_amount","loan_term",
"cibil_score","residential_assets_value","commercial_assets_value","luxury_assets_value",
"bank_asset_value"])

new_data_scaled = scaler.transform(new_data)

prediction = model.predict(new_data_scaled)

result = loan_status_encoder.inverse_transform(prediction)

print("\nLoan Status Prediction:", result[0])
```

Output:

```
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/hp/Desktop/mlai/minipro prediction.py"
Enter number of dependents: 2
Education (Graduate/Not Graduate): Graduate
Self Employed (Yes/No): No
Annual Income: 72000
Loan Amount: 500000
Loan Term: 3
CIBIL Score: 12
Residential Assets Value: 100000
Commercial Assets Value: 120000
Luxury Assets Value: 300000
Bank Asset Value: 25000

Loan Status Prediction: Rejected
PS C:\Users\hp> 
```

Conclusion:

This project successfully developed a loan approval prediction system using machine learning.

The Decision Tree Classifier was trained using applicant information and achieved effective prediction performance. The system can help financial institutions reduce manual effort, improve efficiency, and make faster loan approval decisions. After preprocessing and training, the model achieved an accuracy of 97.78% with a low RMSE value of 0.1491, indicating strong prediction performance and minimal error. The project demonstrated how machine learning can support faster and more efficient decision-making in loan approval processes.

Reflection:

Through this project, I gained a practical understanding of how machine learning can be used to solve real-world problems such as loan approval prediction. I learned the importance of preprocessing steps such as handling missing values, removing duplicate records, encoding categorical variables, and feature scaling to improve model performance.

One of the major challenges faced during implementation was dealing with errors related to dataset column names and categorical values. Some columns contained extra spaces, which caused issues during encoding and prediction. Debugging these errors helped me better understand how datasets should be cleaned and validated before use. The key concepts I learned during this project include data preprocessing, label encoding, train-test split, feature scaling using StandardScaler, model training using Decision Tree Classifier, and model evaluation using accuracy, and RMSE.

Overall, this project helped me understand the complete machine learning workflow, starting from dataset preparation to model building, evaluation, and prediction.